

COMMONWEALTH OF AUSTRALIA

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COMPx270: Randomised and Advanced Algorithms

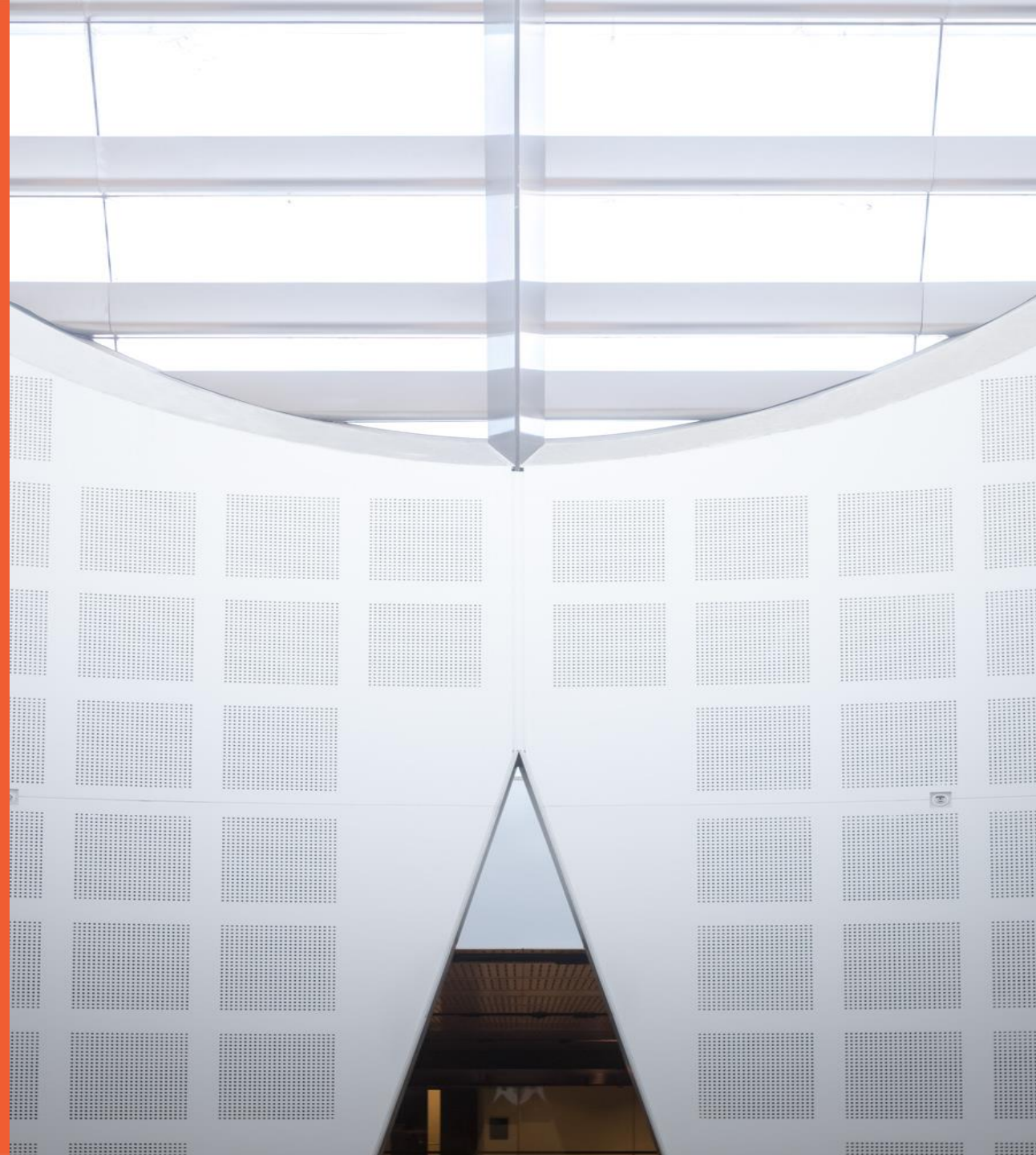
Lecture 7: Nearest Neighbours and dimensionality reduction

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THE UNIVERSITY OF
SYDNEY



A question

You have n pictures, each 4096×4096 pixels, of venomous spiders. Someone finds a spider in their kitchen and sends you a photo, asking which type of spider it is and if it is venomous, **because they just have been bitten.**

How long will it take you?

Nearest Neighbour Search

~~$|X| = m$~~
 ~~X unstructured~~

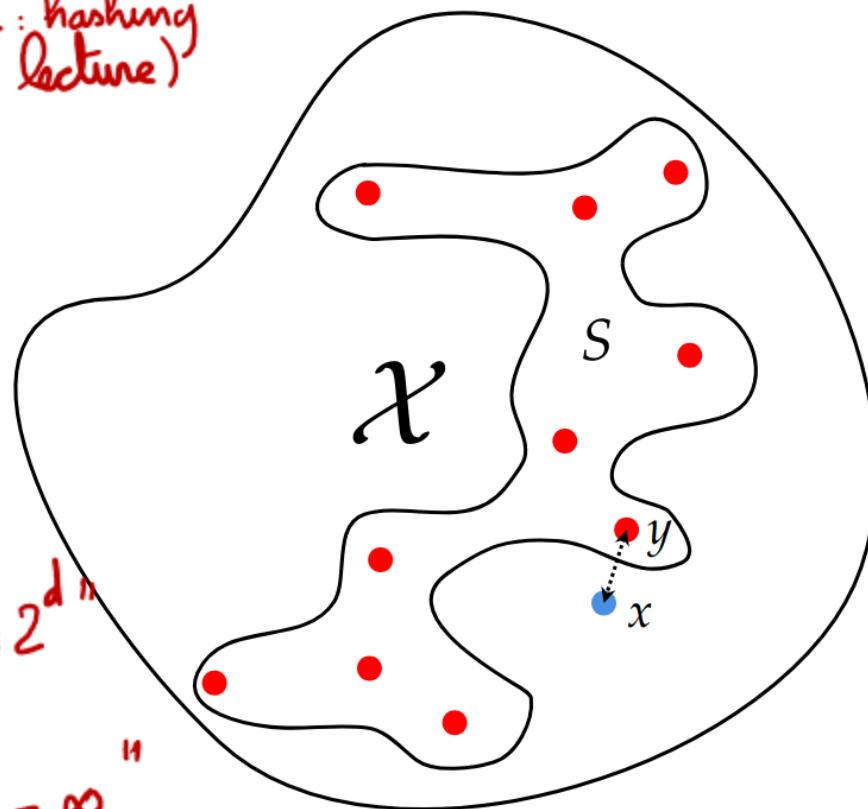
(before: hashing lecture)

(X, dist) metric space

e.g., $X = \{0,1\}^d$ " $m = 2^d$ "

or $X = \mathbb{R}^d$ " $m = \infty$ "

+ $\text{dist} = \text{dist}(x,y)$ tells us how similar x, y are



parameter regime of interest

$$1 \ll d \ll n \ll 2^d$$

Metric

$$\text{dist}: X \times X \rightarrow [0, \infty)$$

$$\textcircled{1} \text{ dist}(x, x) = 0 \quad \forall x$$

$$\textcircled{2} \text{ dist}(x, y) = \text{dist}(y, x)$$

$$\textcircled{3} \text{ dist}(x, y) \leq \text{dist}(x, z) + \text{dist}(z, y)$$

Examples:

* $X = \{0,1\}^d$ $\text{dist}(x,y) = \# \text{ bits where } x, y \text{ differ}$ (Hamming distance)

* $X = \mathbb{R}^d$ $\text{dist}(x,y) = \|x-y\|_2$ (Euclidean)
 or $\text{dist}(x,y) = \|x-y\|_1$ (Manhattan)
 or $\text{dist}(x,y) = \|x-y\|_\infty$ (Infinity)

Nearest Neighbour Search

QUERY(x): on input $x \in \mathcal{X}$, return $y \in S$ such that

$$y = \operatorname{argmin}_{y' \in S} \operatorname{dist}(x, y')$$

dataset of n points
↓

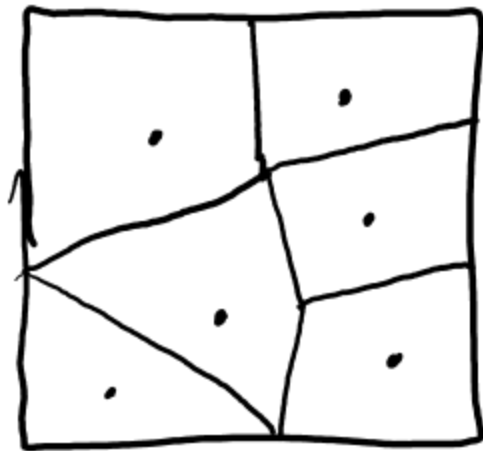
① Query time: ideally $O(1)$, but $O(n)$ is good

② Space: ideally $O(nd)$
of points in S dimension of each point

$O(f)$: at most $c \cdot f$ for some $c > 0$
 $g = o(f)$: less than $c \cdot f$ for all $c > 0$
 $\lim_{n \rightarrow \infty} \frac{g(n)}{f(n)} = 0$

$$1 \ll d \ll n \ll 2^d$$

Lists? Voronoi? K-d trees? Hash tables?



	Query	Space
List	$O(nd)$ ^x	$O(nd)$ [✓]
Array* (for $\{0,1\}^d$)	$O(2^d)$ ^{✓?}	$O(2^d)$ ^x
Voronoi	$O(nd)$ ^x	$O(n^{\lceil d/2 \rceil})$ ^x

~~Hashing~~

Bad news...

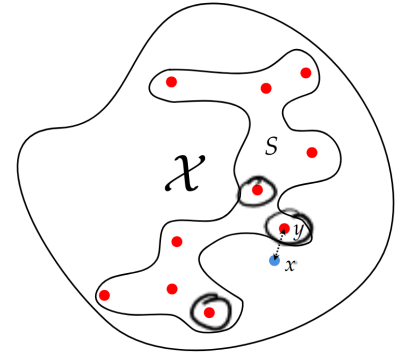
We do not know **anything** (much) better.
Either query time or space is

(NN)

$$\Omega(\min(2^d, nd))$$

for everything we know (even randomized)

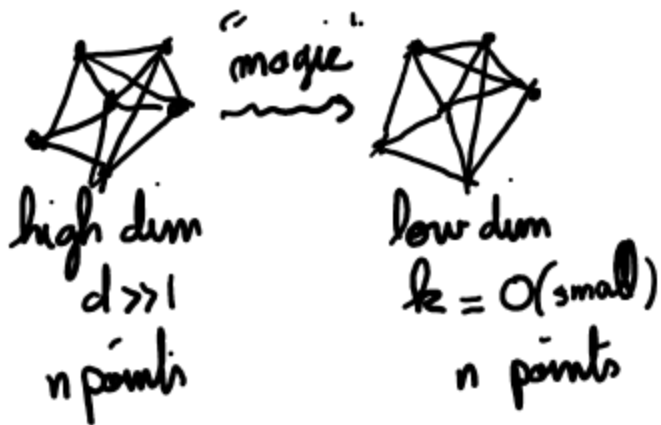
Approximate Nearest Neighbour Search



QUERY(x): on input $x \in \mathcal{X}$, return $y \in S$ sort-of-minimising $\text{dist}(x, y)$, that is, $\text{dist}(x, y) \leq C \cdot \min_{y' \in S} \text{dist}(x, y')$.

$C \geq 1$
($C = 1$ is the NN problem)

Dimensionality Reduction: the JL Lemma (Euclidean space)



$$\Phi: (\mathbb{R}^d, \|\cdot\|_2) \rightarrow (\mathbb{R}^k, \|\cdot\|_2) \quad \text{where } k \ll d$$

Want $\|\Phi(x) - \Phi(y)\|_2 \stackrel{C}{\approx} \|x - y\|_2 \quad \forall x, y \in S$

Theorem (JL Lemma). $\forall \varepsilon > 0$, can do this* with

$$C = 1 + \varepsilon$$

and $k = O\left(\frac{\log n}{\varepsilon^2}\right)$

* Wp 99/100, $(\|\Phi(x) - \Phi(y)\|_2 \in (1 \pm \varepsilon)\|x - y\|_2 \quad \forall x, y \in S)$

and Φ is linear: $\Phi(x) = Ax \in \mathbb{R}^k$

for $A = \frac{1}{\sqrt{k}} \begin{pmatrix} A_{11} & \dots & A_{1d} \\ \vdots & \ddots & \vdots \\ A_{k1} & \dots & A_{kd} \end{pmatrix} \in \mathbb{R}^{k \times d}$ $A_{ij} \sim \mathcal{N}(0, 1)$

Dimensionality Reduction: the JL Lemma (Euclidean space)

We will show a small part: $\forall x, y \in \mathbb{R}^d$ fixed, $\mathbb{E}_A [\|Ax - Ay\|_2^2] = \|x - y\|_2^2$

Suffices* to show: $\forall v \in \mathbb{R}^d$ fixed, $\mathbb{E}_A [\|Av\|_2^2] = \|v\|_2^2$

$\uparrow \Phi(x)$ $\uparrow \Phi(y)$

Suffices* to show: $\forall u \in \mathbb{R}^d$ fixed with $\|u\|_2 = 1$, $\mathbb{E}_A [\|Au\|_2^2] = 1$

*equivalent

Pf. Fix u with $\|u\|_2 = 1$. (†)

$$\begin{aligned} \mathbb{E}_A [\|Au\|_2^2] &= \mathbb{E}_A \left[\sum_{i=1}^k (Au)_i^2 \right] \stackrel{LE}{=} \sum_{i=1}^k \mathbb{E}_A [(Au)_i^2] = \sum_{i=1}^k \mathbb{E}_A \left[\left(\sum_{j=1}^d A_{ij} u_j \right)^2 \right] \\ &= \sum_{i=1}^k \mathbb{E}_A \left[\sum_{j,j'=1}^d A_{ij} A_{ij'} u_j u_{j'} \right] \stackrel{LE}{=} \sum_{i=1}^k \sum_{j,j'=1}^d \underbrace{\mathbb{E}_A [A_{ij} A_{ij'}]}_{\substack{\neq 0 \text{ if } j=j', \\ =0 \text{ if } j \neq j'}} u_j u_{j'} \\ &= \sum_{i=1}^k \sum_{j=1}^d \mathbb{E}_A [A_{ij}^2] u_j^2 = \sum_{i=1}^k \sum_{j=1}^d \left(\frac{1}{\sqrt{k}} \right)^2 \cdot 1 \cdot u_j^2 \quad \left(\substack{A_{ij} = \frac{1}{\sqrt{k}} \times \mathcal{N}(0,1) \\ \mathbb{E}[A_{ij}] = 0, \mathbb{E}[A_{ij}^2] = 1} \right) \\ &= \frac{1}{k} \sum_{i=1}^k \sum_{j=1}^d u_j^2 = 1 \end{aligned}$$

$\sum_{j=1}^d u_j^2 = 1$ (†)

JL Lemma and ANN

QUERY(x): on input $x \in \mathcal{X}$, return $y \in S$ sort-of-minimising $\text{dist}(x, y)$, that is, $\text{dist}(x, y) \leq C \cdot \min_{y' \in S} \text{dist}(x, y')$.

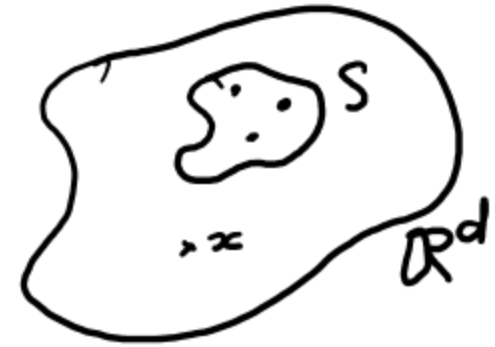
$$\Phi: \mathbb{R}^d \rightarrow \mathbb{R}^k \quad \text{where } k = O\left(\frac{\log(n+1)}{\epsilon^2}\right)$$

Set $C = 1 + \epsilon$

Apply JL to $T = S \cup \{x\}$

PREPARE [Get n points: S
 Compute $\{\Phi(x)\}_{x \in S}: S'$

QUERY [On input $x \in \mathbb{R}^d$ (query)
 - compute $\Phi(x) \in \mathbb{R}^k$
 - Solve NN on $\mathbb{R}^k$ with S'
 (linked list: $O(nk), O(nk)$)
 → gives you ANN on $\mathbb{R}^d$



SPACE $O(nk) = O\left(n \frac{\log n}{\epsilon^2}\right)$

QUERY $O(nk) = O\left(n \frac{\log n}{\epsilon^2}\right)$

Beyond JL Lemma: Hashing!

QUERY(x): on input $x \in \mathcal{X}$, return $y \in S$ sort-of-minimising $\text{dist}(x, y)$, that is, $\text{dist}(x, y) \leq C \cdot \min_{y' \in S} \text{dist}(x, y')$.

Spoiler:

can do

QUERY
(expected)

SPACE

sublinear in n !
This is $o(n)$

$O(n^{\epsilon} d)$ time

$O(n^{1+\epsilon} d)$ ← "mildly" worse than $O(nd)$

for some $\epsilon \in (0, 1)$ (which we don't
"fully" choose)

Hamming: $\epsilon \propto 1/C$

Euclidean: $\epsilon \propto 1/C^2$

Locality-Sensitive Hashing

Definition 36.1. Let $0 \leq q < p \leq 1$, $r > 0$, $C > 1$, and $(\mathcal{X}, \text{dist})$ be a metric space. Then a family of functions $\mathcal{H}$ from $\mathcal{X}$ to $\mathcal{Y}$ is a (r, C, p, q) -Locality Sensitive Hash family (LSH) if, for every $x, x' \in \mathcal{X}$,

- If $\text{dist}(x, x') \leq r$, then $\Pr_{h \sim \mathcal{H}}[h(x) = h(x')] \geq p$;
- If $\text{dist}(x, x') \geq Cr$, then $\Pr_{h \sim \mathcal{H}}[h(x) = h(x')] \leq q$;

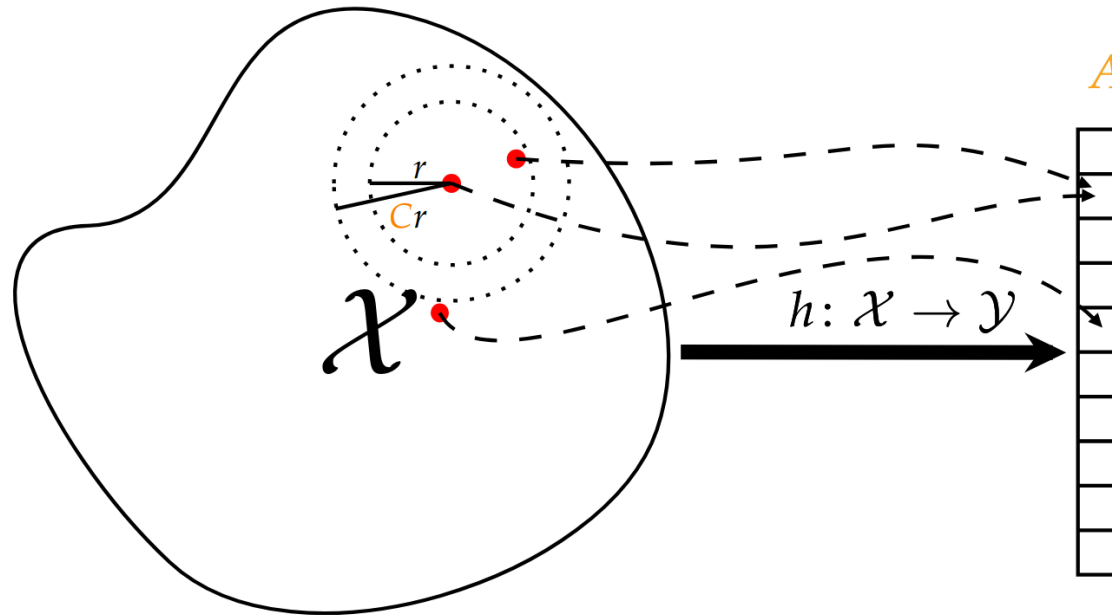
and we say $\rho := \frac{\log(1/p)}{\log(1/q)} < 1$ is the sensitivity parameter of $\mathcal{H}$.

Locality-Sensitive Hashing

Definition 36.1. Let $0 \leq q < p \leq 1$, $r > 0$, $\mathcal{C} > 1$, and $(\mathcal{X}, \text{dist})$ be a metric space. Then a family of functions $\mathcal{H}$ from $\mathcal{X}$ to $\mathcal{Y}$ is a $(r, \mathcal{C}, p, q)$ -Locality Sensitive Hash family (LSH) if, for every $x, x' \in \mathcal{X}$,

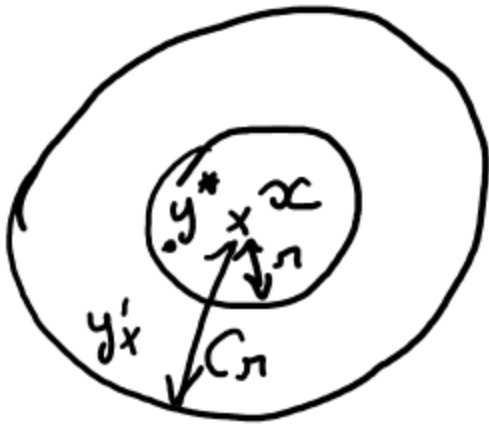
- If $\text{dist}(x, x') \leq r$, then $\Pr_{h \sim \mathcal{H}}[h(x) = h(x')] \geq p$;
- If $\text{dist}(x, x') \geq \mathcal{C}r$, then $\Pr_{h \sim \mathcal{H}}[h(x) = h(x')] \leq q$;

and we say $\rho := \frac{\log(1/p)}{\log(1/q)} < 1$ is the *sensitivity parameter* of $\mathcal{H}$.



Locality-Sensitive Hashing: "Baby version"

π : scale (fixed)



$\text{QUERY}_r(x)$: given an element $x \in \mathcal{X}$, return an element $y \in S$, or $\perp$, such that:

- If there exists $y^* \in S$ such that $\text{dist}(x, y^*) \leq r$, then, with probability at least $9/10$, $\text{QUERY}_r(x)$ returns an element $y \in S$ such that $\text{dist}(x, y^*) \leq C \cdot r$;
- If $\text{dist}(x, y) > C \cdot r$ for every $y \in S$, then, with probability 1, $\text{QUERY}_r(x)$ returns $\perp$.
- Otherwise, any output in $S \cup \{\perp\}$ is allowed.

Locality-Sensitive Hashing: "Baby version" (1/6)

$\text{QUERY}_r(x)$: given an element $x \in \mathcal{X}$, return an element $y \in S$, or $\perp$, such that:

- If there exists $y^* \in S$ such that $\text{dist}(x, y^*) \leq r$, then, with probability at least $9/10$, $\text{QUERY}_r(x)$ returns an element $y \in S$ such that $\text{dist}(x, y^*) \leq c \cdot r$;
- If $\text{dist}(x, y) > c \cdot r$ for every $y \in S$, then, with probability 1, $\text{QUERY}_r(x)$ returns $\perp$.
- Otherwise, any output in $S \cup \{\perp\}$ is allowed.

x is fixed
 C, r is fixed
 $\mathcal{H}$ is given
 $\rightarrow p, q$ are given
 $\rightarrow e = \frac{\log(1/p)}{\log(1/q)}$ is given

Claim From $\mathcal{H}$, can get $\mathcal{H}^{(e)}$
 $(e \geq 1)$ s.t. $\mathcal{H}^{(e)}$ is a (r, C, p^e, q^e) -LSH family (and $|\mathcal{H}^{(e)}| = |\mathcal{H}|^e$)

REDUCE
 "bad"
 q

Pf: $h(x) = (h_1(x), \dots, h_e(x)) \in Y^e$
 where $h_1, \dots, h_e : \mathcal{X} \rightarrow Y$ from $\mathcal{H}$.
 (Proof by case analysis.)

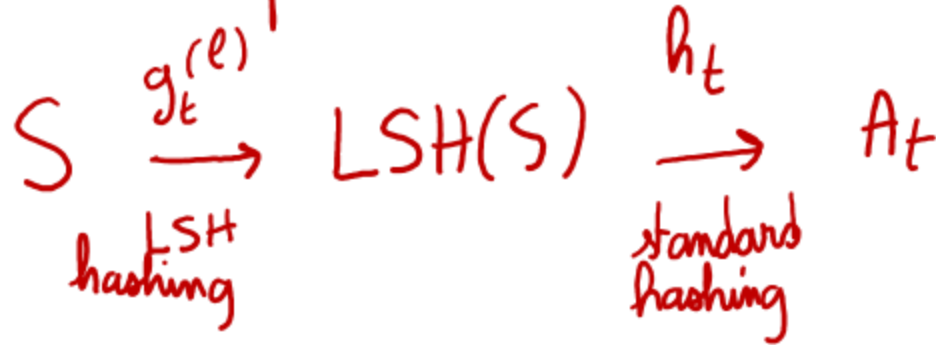
Good! q went down $(q \rightarrow q^e)$
 Bad! p went down $(p \rightarrow q^e)$

Locality-Sensitive Hashing: "Baby version" (2/6)

$\text{QUERY}_r(x)$: given an element $x \in \mathcal{X}$, return an element $y \in S$, or $\perp$, such that:

- If there exists $y^* \in S$ such that $\text{dist}(x, y^*) \leq r$, then, with probability at least $9/10$, $\text{QUERY}_r(x)$ returns an element $y \in S$ such that $\text{dist}(x, y^*) \leq c \cdot r$;
- If $\text{dist}(x, y) > c \cdot r$ for every $y \in S$, then, with probability 1, $\text{QUERY}_r(x)$ returns $\perp$.
- Otherwise, any output in $S \cup \{\perp\}$ is allowed.

Goal: "increase p "



where $1 \leq t \leq k$

Take $[k]$ hash tables $A_1, \dots, A_k$
("standard" hash tables with chaining)

Locality-Sensitive Hashing: "Baby version" (3/6)

$\text{QUERY}_r(x)$: given an element $x \in \mathcal{X}$, return an element $y \in S$, or $\perp$, such that:

- If there exists $y^* \in S$ such that $\text{dist}(x, y^*) \leq r$, then, with probability at least $9/10$, $\text{QUERY}_r(x)$ returns an element $y \in S$ such that $\text{dist}(x, y^*) \leq c \cdot r$;
- If $\text{dist}(x, y) > c \cdot r$ for every $y \in S$, then, with probability 1, $\text{QUERY}_r(x)$ returns $\perp$.
- Otherwise, any output in $S \cup \{\perp\}$ is allowed.

$\text{PREPROCESS}(S)$:

for all $x \in S$ do $\triangleright$ Insert all n elements in all k hash tables

for all $1 \leq t \leq k$ do

$A_t[h_t(g_t(x))].\text{INSERT}(x)$
standard hash $\rightarrow$ *LSH hash*

$\text{QUERY}_r(x)$:

for all $1 \leq t \leq k$ do

$L_t \leftarrow A_t[h_t(g_t(x))]$ $\triangleright$ "Candidates" in t -th hash table

for all $y \in L_t$ do

if $\text{dist}(x, y) \leq c \cdot r$ then

return y

$\triangleright$ Found one!

return $\perp$

$\triangleright$ Did not find any.

Locality-Sensitive Hashing: "Baby version" (4/6)

QUERY_r(x): given an element $x \in \mathcal{X}$, return an element $y \in S$, or $\perp$, such that:

- If there exists $y^* \in S$ such that $\text{dist}(x, y^*) \leq r$, then, with probability at least $9/10$, QUERY_r(x) returns an element $y \in S$ such that $\text{dist}(x, y^*) \leq c \cdot r$;
- If $\text{dist}(x, y) > c \cdot r$ for every $y \in S$, then, with probability 1, QUERY_r(x) returns $\perp$.
- Otherwise, any output in $S \cup \{\perp\}$ is allowed.

Hope • If x, y are close at least one of the k LSH $g_1^{(l)}, \dots, g_k^{(l)}$ will make them collide. When that happens, $y \in L_t$ for that t

→ want $1 - (1 - p^l)^k$ large

- Not too many collisions in the second-level hash tables: it takes query time.

SPACE k hash tables, each n elements of size $O(d)$
 + k LSH^(l) hash functions: $k \times l \times O(d)$
 each of them is l "base" LSH functions $\uparrow$ one from $\mathcal{H}$

$O(knd)$
 $O(kld)$

QUERY TIME ① Evaluate $g_t^l(x) \forall 1 \leq t \leq k$

$O(kl)$
 $O(kdnl)$

② $E\left[\sum_{t=1}^k |L_t| \cdot O(d)\right] \approx k \cdot (nl) \cdot O(d) =$
 $\uparrow$ only need to analyse for "bad collisions"

Locality-Sensitive Hashing: "Baby version" (5/6)

Correctness

When unlucky: there is $y^* \in S$ s.t.

$$\text{dist}(x, y^*) \leq r$$

$$\text{but } g_t^{(l)}(x) \neq g_t^{(l)}(y^*) \quad \forall 1 \leq t \leq k$$

$$\Pr[\text{unlucky}] \leq (1 - p^l)^k \leq \frac{1}{10} \quad \text{WANT} \quad \text{⊛}$$

(Note: we never return a "bad" y , since we check $\text{dist}(x, y) \leq C \cdot r$ in the loop)

How to set k, l ?

to get ⊛ and $nq^l \leq 1$
(correctness) (query time)

$$\text{set } l = O\left(\frac{\log n}{\log(1/q)}\right)$$

$$\text{and } k = \Theta(n^l)$$

$$\text{Space: } O(knd) = O(n^{l+d})$$

$$E[\text{query time}] = O(kd n q^l) = O(n^l d)$$

$\text{QUERY}_r(x)$: given an element $x \in \mathcal{X}$, return an element $y \in S$, or $\perp$, such that:

- If there exists $y^* \in S$ such that $\text{dist}(x, y^*) \leq r$, then, with probability at least $9/10$, $\text{QUERY}_r(x)$ returns an element $y \in S$ such that $\text{dist}(x, y^*) \leq C \cdot r$;
- If $\text{dist}(x, y) > C \cdot r$ for every $y \in S$, then, with probability 1, $\text{QUERY}_r(x)$ returns $\perp$.
- Otherwise, any output in $S \cup \{\perp\}$ is allowed.

Unlucky: "does not return a good y even though there is one"

IN ANALYSIS HERE, ONLY FOCUS ON COLLISIONS FROM LSH STAGE

(we also have "regular hash table" collisions to take into account, but we know how to quantify them from last week
→ let's not worry about them for now)

Locality-Sensitive Hashing: "Baby version" (6/6)

$\text{QUERY}_r(x)$: given an element $x \in \mathcal{X}$, return an element $y \in S$, or $\perp$, such that:

- If there exists $y^* \in S$ such that $\text{dist}(x, y^*) \leq r$, then, with probability at least $9/10$, $\text{QUERY}_r(x)$ returns an element $y \in S$ such that $\text{dist}(x, y^*) \leq c \cdot r$;
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Theorem. *With suitable settings of k, ℓ and assuming $q, T = \Theta(1)$, the data structure for the “baby version of ANN” has space complexity $O(n^{1+\rho} d)$ and expected query complexity $O(n^\rho d)$.*

Locality-Sensitive Hashing: "Baby version" (👶)

But how to solve ANN?

There is no forced "0"
or option to output "1"
in the ANN task.

Locality-Sensitive Hashing: "They grow up so fast" (👶)

"Binary search
even π "

Loses $\log d$ factors.

(conditions
apply)

For instance, Hamming space:

$$0 \leq \underbrace{\text{dist}(x, y)}_{\text{integer}} \leq d$$

so can binary search for π .

Locality-Sensitive Hashing: But... do they exist?

Yes!

Example

Hamming: $\{0,1\}^d$

Choose $h_i(x) = x_i \in \{0,1\}$, $\mathcal{H} = \{h_i : i \in [d]\}$

Given C, η :

If $\text{dist}(x, x') \leq \eta$

$$\Pr_{h \sim \mathcal{H}} [h(x) = h(x')] \geq 1 - \frac{\eta}{d} \triangleq p$$

If $\text{dist}(x, x') \geq C \cdot \eta$

$$\Pr_{h \sim \mathcal{H}} [h(x) = h(x')] \leq 1 - \frac{C\eta}{d} \triangleq q$$

$$e = \frac{\log(1 - \frac{\eta}{d})}{\log(1 - \frac{C\eta}{d})} \stackrel{\text{can be shown}}{\approx} \frac{1}{C}$$

Locality-Sensitive Hashing: But... do they exist?

Also exist for $(\mathbb{R}^d, \ell_2)$, and many
more.

See tutorial for some!